PHYSICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

26. The workdone in an adiabatic change in an ideal gas depends upon only :
    (1) Change in its volume
    (2) Change in its temperature
    (3) Change in its pressure
    (4) Change in its specific heat

Answer (2)

Sol. Work done in adiabatic process = $\frac{nR\Delta T}{1-\gamma}$

So, depends upon change in temperature.

27. Two projectiles are fired with same initial speed from same point on ground at angles of $(45° - \alpha)$ and $(45° + \alpha)$, respectively, with the horizontal direction. The ratio of their maximum heights attained is :
    (1) $\frac{1+\sin 2\alpha}{1-\sin 2\alpha}$ (2) $\frac{1+\sin\alpha}{1-\sin\alpha}$
    (3) $\frac{1-\tan\alpha}{1+\tan\alpha}$ (4) $\frac{1-\sin 2\alpha}{1+\sin 2\alpha}$

Answer (4)

Sol. $H_{max} = \frac{u^2\sin^2\theta}{2g}$

$$\frac{H_1}{H_2} = \frac{\sin^2(45-\alpha)}{\sin^2(45+\alpha)}$$

$$= \frac{(\cos\alpha-\sin\alpha)^2}{(\cos\alpha+\sin\alpha)^2}$$

$$\frac{H_1}{H_2} = \frac{1-\sin 2\alpha}{1+\sin 2\alpha}$$

28. Consider a long straight wire of a circular cross-section (radius *a*) carrying a steady current *I*. The current is uniformly distributed across this cross-section. The distances from the centre of the wire's cross-section at which the magnetic field [inside the wire, outside the wire] is half of the maximum possible magnetic field, anywhere due to the wire, will be
    (1) [a/2, 2a] (2) [a/2, 3a]
    (3) [a/4, 2a] (4) [a/4, 3a/2]

Answer (1)

Sol. Magnetic field inside cylinder at distance *r* from axis

$$= \frac{\mu_0 I}{\pi R^2}\cdot\frac{r}{2}$$

$$B_{max} = \frac{\mu_0 I}{2\pi R}$$

$$\frac{\mu_0 I}{2\pi R^2} r_1 = \frac{1}{2}\left(\frac{\mu_0 I}{2\pi R}\right)$$

$$r_1 = \frac{R}{2} = \frac{a}{2}$$

Magnetic field outside cylinder at distance *r* from axis

$$= \frac{\mu_0 I}{2\pi r}$$

$$\frac{\mu_0 I}{2\pi r_2} = \frac{1}{2}\left(\frac{\mu_0 I}{2\pi R}\right)$$

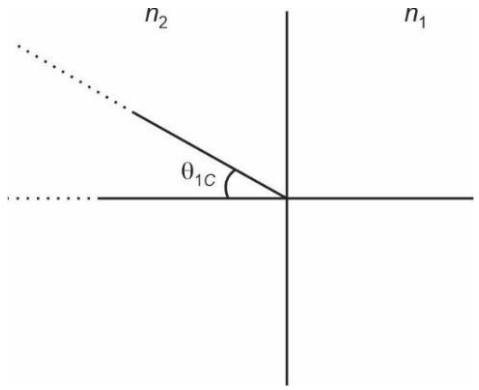
$r_2 = 2R = 2a$

29. At the interface between two materials having refractive indices n_1 and n_2, the critical angle for reflection of an em wave is θ_{1C}. The n_2 material is replaced by another material having refractive index n_3 such that the critical angle at the interface between n_1 and n_3 materials is θ_{2C}. If $n_3 > n_2 > n_1$; $\frac{n_2}{n_3} = \frac{2}{5}$ and $\sin\theta_{2C} - \sin\theta_{1C} = \frac{1}{2}$, then θ_{1C} is
    (1) $\sin^{-1}\left(\frac{1}{3n_1}\right)$ (2) $\sin^{-1}\left(\frac{1}{6n_1}\right)$
    (3) $\sin^{-1}\left(\frac{5}{6n_1}\right)$ (4) $\sin^{-1}\left(\frac{2}{3n_1}\right)$

Answer (*)

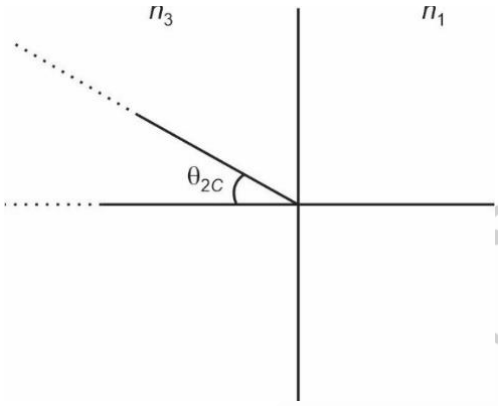


Sol.



$$n_2 \sin(\theta_{1C}) = n_1$$

$$\sin(\theta_{1C}) = \frac{n_1}{n_2}$$



$$n_3 \sin(\theta_{2C}) = n_1$$

$$\Rightarrow \quad \sin(\theta_{2C}) = \frac{n_1}{n_3}$$

Also, $n_3 = \frac{5n_2}{2} \Rightarrow \sin(\theta_{2C}) = \frac{2n_1}{5n_2}$

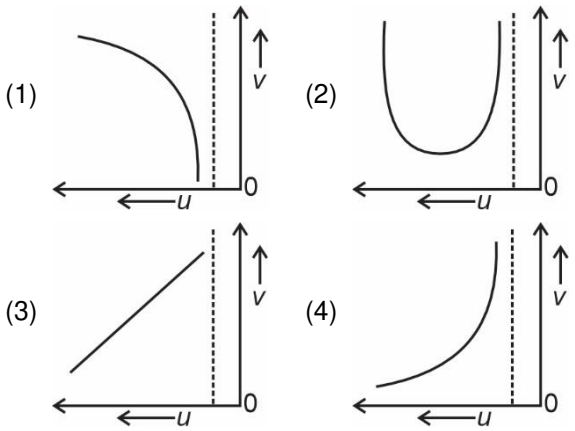
$$\Rightarrow \quad \frac{n_1}{n_2} = \frac{5}{2}.\sin(\theta_{2C}) \Rightarrow \sin(\theta_{2C}) - \sin(\theta_{1c}) = \frac{1}{2}$$

Given, $\frac{2n_1}{5n_2} - \frac{n_1}{n_2} = \frac{1}{2} \Rightarrow \frac{n_1}{n_2}\left(\frac{-3}{5}\right) = \frac{1}{2}$

Coming out to be (–ve)

*** None of the answer is matching**

30. Let u and v be the distances of the object and the image from a lens of focal length f. The correct graphical representation of u and v for a convex lens when $|u| > f$, is



Answer (4)

Sol. Lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u}$$

Since $|\mu| > f$

So, RHS is positive.

31. The pair of physical quantities not having same dimensions is :

(1) Angular momentum and Planck's constant

(2) Torque and energy

(3) Surface tension and impulse

(4) Pressure and Young's modulus

Answer (3)

Sol. [Angular momentum] = ML^2T^{-1}

[Planck's Constant] = ML^2T^{-1}

[Torque] = ML^2T^{-2}

[Energy] = ML^2T^{-2}

[Surface tension] = MT^{-2}

[Impulse]= MLT^{-1}

[Pressure] = $ML^{-1}T^{-2}$

[Young's modulus] = $ML^{-1}T^{-2}$



32. The expression given below shows the variation of velocity (v) with lime (t), $v = At^2 + \frac{Bt}{C+t}$. The dimension of ABC is :

(1) $[M^0L^1T^{-2}]$ (2) $[M^0L^1T^{-3}]$

(3) $[M^0L^2T^{-3}]$ (4) $[M^0L^2T^{-2}]$

Answer (3)

Sol. $v = At^2 + \frac{Bt}{C+t}$

$[v] = [A]\, t^2 = \left[\frac{Bt}{C+t}\right]$

$[A] = LT^{-3}$

$[C] = T$

$[B] = LT^{-1}$

$[ABC] = L^2T^{-3}$

33. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) : Choke coil is simply a coil having a large inductance but a small resistance. Choke coils are used with fluorescent mercury-tube fittings. If household electric power is directly connected to a mercury tube, the tube will be damaged.

Reason (R) : By using the choke coil, the voltage across the tube is reduced by a factor $\left(R/\sqrt{R^2+\omega^2L^2}\right)$, where ω is frequency of the supply across resistor R and inductor L. If the choke coil were not used, the voltage across the resistor would be the same as the applied voltage.

In the light of the above statements, choose the **most appropriate answer** from the options given below:

(1) Both **(A)** and **(R)** are true but **(R)** is not the correct explanation of **(A)**

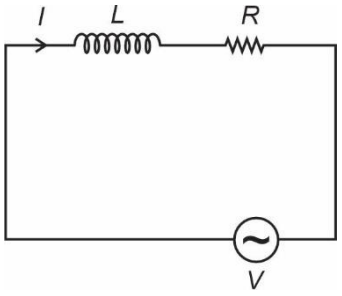
(2) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of (A)

(3) **(A)** is false but **(R)** is true

(4) **(A)** is true but **(R)** is false

Answer (1)

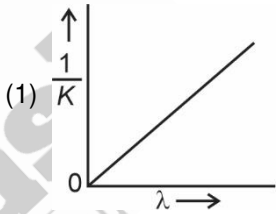
Sol.

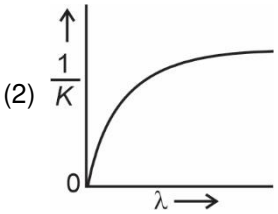


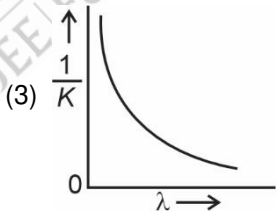
$I = \frac{V}{\sqrt{R^2+\omega^2L^2}}$

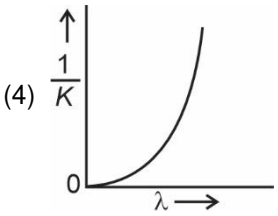
$V_R = \frac{R}{\sqrt{R^2+\omega^2L^2}}V$

34. If λ and K are de Broglie wavelength and kinetic energy, respectively, of a particle with constant mass. The correct graphical representation for the particle will be

(1) 

(2) 

(3) 

(4) 

Answer (4)

Sol. $\lambda = \frac{h}{\sqrt{2mK}}$

$\lambda^2 = \frac{h^2}{2mK}$

$\frac{1}{K} = \left(\frac{2m}{h^2}\right)\lambda^2$



35. Consider I_1 and I_2 are the currents flowing simultaneously in two nearby coils 1 & 2, respectively. If L_1 = sell inductance of coil 1, M_{12} = mutual inductance of coil 1 with respect to coil 2, then the value of induced emf in coil 1 will be

(1) $\varepsilon_1 = -L_1\frac{dI_1}{dt} - M_{12}\frac{dI_2}{dt}$

(2) $\varepsilon_1 = -L_1\frac{dI_2}{dt} - M_{12}\frac{dI_1}{dt}$

(3) $\varepsilon_1 = -L_1\frac{dI_1}{dt} + M_{12}\frac{dI_2}{dt}$

(4) $\varepsilon_1 = -L_1\frac{dI_1}{dt} - M_{12}\frac{dI_1}{dt}$

Answer (1)

Sol. Magnitude of induced emf due to self inductance

$= \frac{LdI_1}{dt}$

Magnitude of induced emf due to mutual inductance

$= \frac{MdI_2}{dt}$

36. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) : Electromagnetic waves carry energy but not momentum.

Reason (R) : Mass of a photon is zero.

In the light of the above statements, choose the **most appropriate answer** from the options given below :

(1) Both **(A)** and **(R)** are true but **(R)** is not the correct explanation of **(A)**

(2) **(A)** is false but **(R)** is true

(3) **(A)** is true but **(R)** is false

(4) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**

Answer (2)

Sol. EM wave carry both energy and momentum. Rest mass of photon is zero.

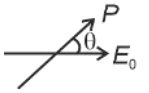
37 An electric dipole of mass m, charge q, and length l is placed in a uniform electric field $\vec{E} = E_0\hat{i}$. When the dipole is rotated slightly from its equilibrium position and released, the time period of its oscillations will be :

(1) $2\pi\sqrt{\frac{ml}{2qE_0}}$ (2) $\frac{1}{2\pi}\sqrt{\frac{ml}{2qE_0}}$

(3) $2\pi\sqrt{\frac{ml}{qE_0}}$ (4) $\frac{1}{2\pi}\sqrt{\frac{2ml}{qE_0}}$

Answer (1)

Sol. $\longrightarrow E_0$



$\tau = PE_0\sin\theta$

If θ is small

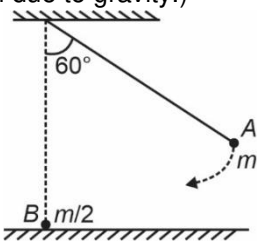
$\tau = -(PE_0)\theta$

$I = m\left(\frac{l}{2}\right)^2\cdot 2 = \frac{ml^2}{2}$

$T = 2\pi\sqrt{\frac{ml^2}{2\cdot PE_0}} = 2\pi\sqrt{\frac{ml^2}{2\cdot qlE_0}}$

$T = 2\pi\sqrt{\frac{ml}{2qE_0}}$

38. As shown below, bob A of a pendulum having massless string of length 'R' is released from 60° to the vertical. It hits another bob B of half the mass that is at rest on a frictionless table in the center. Assuming elastic collision, the magnitude of the velocity of bob A after the collision will be (take g as acceleration due to gravity.)



(1) $\frac{1}{3}\sqrt{Rg}$ (2) $\sqrt{Rg}$

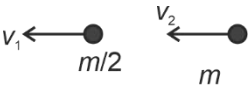
(3) $\frac{4}{3}\sqrt{Rg}$ (4) $\frac{2}{3}\sqrt{Rg}$

Answer (1)



Sol. Velocity of A before collision $= \sqrt{2gh}$

$$= \sqrt{2g \times \frac{R}{2}} = \sqrt{Rg}$$

After collision 

COM

$$mu = \frac{m}{2}v_1 + mv_2$$

$2u = v_1 + 2v_2$ …(i)

$e = 1,\ u = v_1 - v_2$ …(ii)

$u = 3v_2$

$$v_2 = \frac{u}{3} = \frac{1}{3}\sqrt{Rg}$$

39. Match **List-I** with **List-II**.

	List-I		List-II
(A)	Electric field inside (distance $r > 0$ from center) of a uniformly charged spherical shell with surface charge density σ, and radius R.	(I)	σ / ε_0
(B)	Electric field at distance $r > 0$ from a uniformly charged infinite plane sheet with surface charge density σ.	(II)	$\sigma / 2\varepsilon_0$
(C)	Electric field outside (distance $r > 0$ from center) of a uniformly charged spherical shell with surface charge density σ, and radius R.	(III)	0
(D)	Electric field between 2 oppositely charged infinite plane parallel sheets with uniform surface charge density σ.	(IV)	$\frac{\sigma}{\varepsilon_0 r^2}$

Choose the **correct** answer from the options given below:

(1) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

(2) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

(3) (A)-(IV), (B)-(II), (C)-(III), (D)-(I)

(4) (A)-(IV), (B)-(I), (C)-(III), (D)-(II)

Answer (*)

Sol. Inside uniformly charged spherical shell, $E = 0$

$\therefore$ A $\rightarrow$ III

For uniformly charged infinite plate

$$E = \frac{\sigma}{2\varepsilon_0}$$

B $\rightarrow$ II

Outside of spherical shell

$$E = \frac{Q}{4\pi\varepsilon_0 r_2} = \frac{\sigma R^2}{\varepsilon_0 r^2}$$

None of the option is matching for C.

Between two plates $E = \frac{\sigma}{\varepsilon_0}$

D $\rightarrow$ I

None of the option is correct

40. The fractional compression $\left(\frac{\Delta V}{V}\right)$ of water at the depth of 2.5 km below the sea level is _____ %. Given, the Bulk modulus of water = 2×10^9 Nm^{-2}, density of water = 10^3 kg m^{-3}, acceleration due to gravity = $g = 10$ ms^{-2}.

(1) 1.25

(2) 1.5

(3) 1.0

(4) 1.75

Answer (1)

Sol. $B = \frac{\Delta P}{-\left(\frac{\Delta V}{V}\right)}$

$$-\left(\frac{\Delta V}{V}\right) = \frac{\Delta P}{B} = \frac{\rho g h}{B}$$

$$= \frac{10^3 \times 10 \times 2.5 \times 10^3}{2 \times 10^9} = 1.25\%$$



41. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) : Emission of electrons in photoelectric effect can be suppressed by applying a sufficiently negative electron potential to the photoemissive substance.

Reason (R) : A negative electric potential, which stops the emission of electrons from the surface of a photoemissive substance, varies linearly with frequency of incident radiation.

In the light of the above statements, choose the **most appropriate answer** from the options given below :

(1) **(A)** is true but **(R)** is false

(2) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**

(3) **(A)** is false but **(R)** is true

(4) Both **(A)** and **(R)** are true but **(R)** is **not** the correct explanation of **(A)**

Answer (4)

Sol. Negative potential will slow the electrons and if it is sufficient, it will make the photocurrent zero.

$eVs = hf - \phi_0$

42. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) : Time period of a simple pendulum is longer at the top of a mountain than that at the base of the mountain.

Reason (R) : Time period of a simple pendulum decreases with increasing value of acceleration due to gravity and vice-versa.

In the light of the above statements, choose the **most appropriate answer** from the options given below :

(1) **(A)** is false but **(R)** is true

(2) **(A)** is true but **(R)** is false

(3) Both **(A)** and **(R)** are true but **(R)** is not the correct explanation of **(A)**

(4) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**

Answer (4)

Sol. $T = 2\pi\sqrt{\frac{l}{g}}$

At top of mountain $g\downarrow$, $\therefore T\uparrow$

43. A coil of area A and N turns is rotating with angular velocity ω in a uniform magnetic field $\vec{B}$ about an axis perpendicular to $\vec{B}$. Magnetic flux φ and induced emf ε across it, at an instant when $\vec{B}$ is parallel to the plane of coil, are :

(1) $\varphi = 0, \varepsilon = 0$

(2) $\varphi = AB, \varepsilon = 0$

(3) $\varphi = AB, \varepsilon = NAB\omega$

(4) $\varphi = 0, \varepsilon = NAB\omega$

Answer (3)

Sol. $\phi = NBA\cos\theta$

$$\varepsilon = -\frac{d\phi}{dt} = -NBA\frac{d\cos\theta}{dt}$$

$\theta = \omega t$

$\varepsilon = -NBA\omega\sin\omega t$

if B is parallel to plane of coil

$\theta = 90°$

$\phi = 0$, $E = BA\omega N$

44. A body of mass 'm' connected to a massless and unstretchable string goes in verticle circle of radius 'R' under gravity g. The other end of the string is fixed at the center of circle. If velocity at top of circular path is $n\sqrt{gR}$, where, $n \geq 1$, then ratio of kinetic energy of the body at bottom to that at top of the circle is

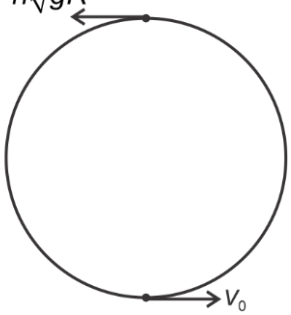
(1) $\frac{n^2+4}{n^2}$ (2) $\frac{n}{n+4}$

(3) $\frac{n+4}{n}$ (4) $\frac{n^2}{n^2+4}$

Answer (1)



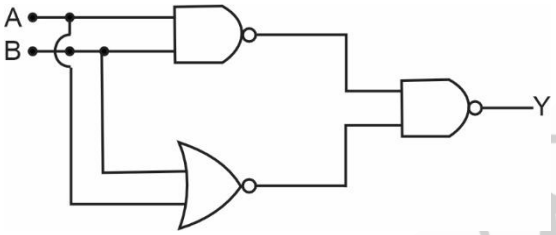
Sol. $v = n\sqrt{gR}$



$$v_0 = \sqrt{v^2 + 2g(2R)}$$

$$v_0 = \sqrt{n^2 gR + 4gR}$$

$$\therefore \quad \frac{k_{bottom}}{k_{top}} = \frac{v_0^2}{v^2} = \frac{n^2+4}{n^2}$$

45. 

For the circuit shown above, equivalent GATE is :

(1) AND gate

(2) OR gate

(3) NOT gate

(4) NAND gate

Answer (2)

Sol. $Y = \overline{AB\left(\overline{A+B}\right)}$

$= AB + A + B$

$= A(B+1) + B$

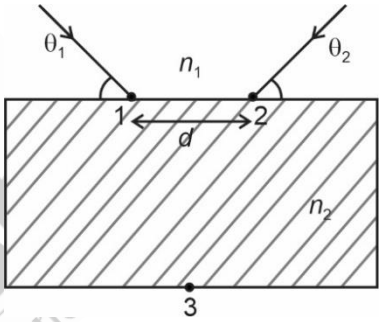
$= A + B$

or GATE

SECTION - B

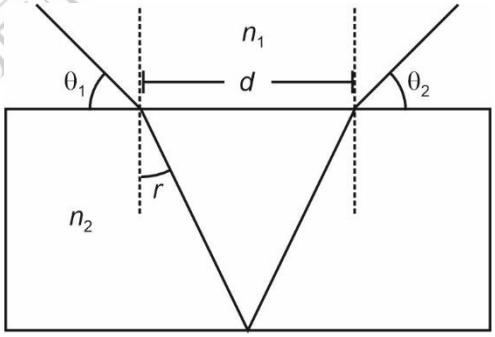
Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

46. Two light beams fall on a transparent material block at point 1 and 2 with angle θ_1 and θ_2, respectively, as shown in figure. After refraction, the beams intersect at point 3 which is exactly on the interface at other end of the block. Given : the distance between 1 and 2, $d = 4\sqrt{3}$ cm and $\theta_1 = \theta_2 = \cos^{-1}\left(\frac{n_2}{2n_1}\right)$, where refractive index of the block n_2 > refractive index of the outside medium n_1, then the thickness of the block is _____ cm.



Answer (6)

Sol.



$$n_1 \sin(90 - \theta_1) = n_2 \sin r$$

$$n_1 \times \frac{n_2}{2n_1} = n_2 \sin r$$

$$\sin r = \frac{1}{2}$$



$r = 30°$

$$\tan r = \left(\frac{d/2}{t}\right)$$

$$t = \frac{d}{2\tan r} = \frac{d\sqrt{3}}{2} = \frac{\left(4\sqrt{3}\right)\sqrt{3}}{2}$$

$= 6$ cm

47. A container of fixed volume contains a gas at 27° C. To double the pressure of the gas, the temperature of gas should be raised to ____ °C.

Answer (327)

Sol. V = constant

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

$P_2 = 2P_1$

$T_2 = 2T_1$

$= 2 \times 300 = 600$ K

$\therefore$ $= 327°C$

48. The maximum speed of a boat in still water is 27 km/h. Now this boat is moving downstream in a river flowing at 9 km/h. A man in the boat throws a ball vertically upwards with speed of 10 m/ s. Range of the ball as observed by an observer at rest on the river bank, is _____ cm. (Take $g = 10$ m/s^2)

Answer (2000)

Sol. $v_y = 10$ m/s

$v_x = v_{river} + v_{boat} = 27 + 9$

$= 36$ km/h $= 10$ m/s

$$R = \left(\frac{2v_y}{g}\right)v_x = \frac{2\times10\times10}{10} = 20 \text{ m}$$

$= 2000$ cm

49. In a hydraulic lift, the surface area of the input piston is 6 cm^2 and that of the output piston is 1500 cm^2. If 100 N force is applied to the input piston to raise the output piston by 20 cm, then the work done is _____ kJ.

Answer (5)

Sol. According to Pascal's law

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$

$$F_2 = \left(\frac{A_2}{A_1}\right)F_1 = \frac{1500}{6}\times100\text{N}$$

$$W = F_2 d_2 = \frac{1500}{6}\times100\times\frac{20}{100}$$

$= 5$ kJ

50. The coordinates of a particle with respect to origin in a given reference frame is (1, 1, 1) meters. If a force of $\vec{F} = \hat{i} - \hat{j} + \hat{k}$ acts on the particle, then the magnitude of torque (with respect to origin) in z-direction is _____.

Answer (2)

Sol. $\tau = \vec{r} \times \vec{F}$

$$\tau = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 1 & -1 & 1 \end{vmatrix}$$

Along $\hat{k}$

$\tau_2 = -2\hat{k}$



CHEMISTRY

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

51. For a $Mg|Mg^{2+}(aq) \,||\, Ag^{+}(aq) \,|\, Ag$ the correct Nernst equation is :

(1) $E_{cell} = E^{\circ}_{cell} + \frac{RT}{2F}\ln\frac{[Ag^{+}]^2}{[Mg^{2+}]}$

(2) $E_{cell} = E^{\circ}_{cell} - \frac{RT}{2F}\ln\frac{[Ag^{+}]^2}{[Mg^{2+}]}$

(3) $E_{cell} = E^{\circ}_{cell} - \frac{RT}{2F}\ln\frac{[Ag^{+}]}{[Mg^{2+}]}$

(4) $E_{cell} = E^{\circ}_{cell} - \frac{RT}{2F}\ln\frac{[Mg^{2+}]}{[Ag^{+}]}$

Answer (1)

Sol.

Cathode	$(Ag^{+}(aq) + e^{-} \rightarrow Ag) \times 2$	
Anode	Mg	$\rightarrow Mg^{2+}(aq) + 2e^{-}$
Cell reaction	$2Ag^{+} + Mg$	$\rightarrow 2Ag + Mg^{2+}$

$$Q = \frac{[Mg^{2+}]}{[Ag^{+}]^2}$$

By Nernst equation

$$E_{cell} = E^{o}_{cell} - \frac{RT}{nF}\ln Q$$

$$E_{cell} = E^{o}_{cell} - \frac{RT}{nF}\ln\frac{[Mg^{2+}]}{[Ag^{+}]^2}$$

$$= E^{o}_{cell} + \frac{RT}{2F}\ln\frac{[Ag^{+}]^2}{[Mg^{2+}]}$$

52. 1.24 g of AX_2 (molar mass 124 g mol^{-1}) is dissolved in 1 kg of water to form a solution with boiling point of 100.0156°C, while 25.4 g of AY_2 (molar mass 250 g mol^{-1}) in 2 kg of water constitutes a solution with a boiling point of 100.0260°C.

$K_b(H_2O)$ = 0.52 K kg mol^{-1}

Which of the following is **correct** ?

(1) AX_2 and AY_2 (both) are fully ionised

(2) AX_2 is fully ionised while AY_2 is completely unionised

(3) AX_2 and AY_2 (both) are completely unionised

(4) AX_2 is completely unionised while AY_2 is fully ionised

Answer (2)

Sol. For AX_2

$\Delta T_b = iK_bm$

$$0.0156 = i \times 0.52 \times \frac{1.24}{124 \times 1}$$

$3 = i$

$3 = 1 + 2\alpha$

$1 = \alpha$

For AY_2

$\Delta T_b = iK_bm$

$$0.0260 = i \times 0.52 \times \frac{25.4}{250 \times 2}$$

$i \approx 1$

$\therefore$ AX_2 is completely ionised & AY_2 is completely unionised

